

STOCK MAINTENANCE SQL QUERIES

Database and table creation:

```
create database stock_maintenance;  
use stock_maintenance;
```

✓	1	10:50:25	create database stock_maintenance	1 row(s) affected
✓	2	10:50:37	use stock_maintenance	0 row(s) affected

```
create table stock(product_id int primary key not null, product_name varchar(250) not null, category varchar(100) not null,  
quantity int not null, price int not null, supplier varchar(250) not null);
```

✓	3	10:54:24	create table stock(product_id int primary key not null, product_name varchar(250) not null, category varchar(100) ...	0 row(s) affected
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Inserting values:

```
5 • insert into stock values(1, "Laptop", "Electronics",1, 2402900, "ABC Electronics");  
6 • insert into stock values(2, "Mobile phone","Electronics", 2, 23920, "Oppo");  
7 • insert into stock values(3, "Portable Mixer", "Home appliances", 4, 2239, "Suji appliances");  
8 • insert into stock values(4, "Short kurti", "Women's dress", 10, 9829, "Pappu Readymades");  
9 • insert into stock values(5, "Chair", "Home appliances", 8, 82749,"Suji appliances");  
10 • insert into stock values(6, "Fan","Home appliances", 7, 23245,"Suji appliances");  
11 • insert into stock values(7, "Jeans","Women dress",4, 64728, "Pappu Readymades");  
12 • insert into stock values(8, "Watch","Accessories", 10, 75296,"Watch Mechanics");  
13 • insert into stock values(9, "Laptop table", "Home appliances",9,862263, "Suji appliances");  
14 • insert into stock values(10, "Extension box", "Home appliances", 7, 975826,"Suji appliances");
```

✓	4	11:05:22	insert into stock values(1, "Laptop", "Electronics",1, 2402900, "ABC Electronics")	1 row(s) affected
✓	5	11:05:27	insert into stock values(2, "Mobile phone","Electronics", 2, 23920, "Oppo")	1 row(s) affected
✓	6	11:05:27	insert into stock values(3, "Portable Mixer", "Home appliances", 4, 2239, "Suji appliances")	1 row(s) affected
✓	7	11:05:27	insert into stock values(4, "Short kurti", "Women's dress", 10, 9829, "Pappu Readymades")	1 row(s) affected
✓	8	11:05:27	insert into stock values(5, "Chair", "Home appliances", 8, 82749,"Suji appliances")	1 row(s) affected
✓	9	11:05:27	insert into stock values(6, "Fan","Home appliances", 7, 23245,"Suji appliances")	1 row(s) affected
✓	10	11:05:27	insert into stock values(7, "Jeans","Women dress",4, 64728, "Pappu Readymades")	1 row(s) affected
✓	11	11:05:27	insert into stock values(8, "Watch","Accessories", 10, 75296,"Watch Mechanics")	1 row(s) affected
✓	12	11:05:27	insert into stock values(9, "Laptop table", "Home appliances",9,862263, "Suji appliances")	1 row(s) affected
✓	13	11:05:27	insert into stock values(10, "Extension box", "Home appliances", 7, 975826,"Suji appliances")	1 row(s) affected

1.Select and where:

1.1 Displaying all products from stock:

```
select * from stock;
```

	product_id	product_name	category	quantity	price	supplier
▶	1	Laptop	Electronics	1	2402900	ABC Electronics
	2	Mobile phone	Electronics	2	23920	Oppo
	3	Portable Mixer	Home appliances	4	2239	Suji appliances
	4	Short kurti	Women's dress	10	9829	Pappu Readymades
	5	Chair	Home appliances	8	82749	Suji appliances
	6	Fan	Home appliances	7	23245	Suji appliances
	7	Jeans	Women dress	4	64728	Pappu Readymades
	8	Watch	Accessories	10	75296	Watch Mechanics
	9	Laptop table	Home appliances	9	862263	Suji appliances
	10	Extension box	Home appliances	7	975826	Suji appliances
*	NULL	NULL	NULL	NULL	NULL	NULL

1.2 Display the product name, category, and price of all products:

```
select product_name, category, price from stock;
```

	product_name	category	price
▶	Laptop	Electronics	2402900
	Mobile phone	Electronics	23920
	Portable Mixer	Home appliances	2239
	Short kurti	Women's dress	9829
	Chair	Home appliances	82749
	Fan	Home appliances	23245
	Jeans	Women dress	64728
	Watch	Accessories	75296
	Laptop table	Home appliances	862263
	Extension box	Home appliances	975826

1.3 Display products whose quantity is less than 10.

```
select * from stock where quantity<10;
```

	product_id	product_name	category	quantity	price	supplier
▶	1	Laptop	Electronics	1	2402900	ABC Electronics
	2	Mobile phone	Electronics	2	23920	Oppo
	3	Portable Mixer	Home appliances	4	2239	Suji appliances
	5	Chair	Home appliances	8	82749	Suji appliances
	6	Fan	Home appliances	7	23245	Suji appliances
	7	Jeans	Women dress	4	64728	Pappu Readymades
	9	Laptop table	Home appliances	9	862263	Suji appliances
	10	Extension box	Home appliances	7	975826	Suji appliances
*	NULL	NULL	NULL	NULL	NULL	NULL

1.4 Display products whose price is greater than 1000.

```
select * from stock where price>1000;
```

	product_id	product_name	category	quantity	price	supplier
▶	1	Laptop	Electronics	1	2402900	ABC Electronics
	2	Mobile phone	Electronics	2	23920	Oppo
	3	Portable Mixer	Home appliances	4	2239	Suji appliances
	4	Short kurti	Women's dress	10	9829	Pappu Readymades
	5	Chair	Home appliances	8	82749	Suji appliances
	6	Fan	Home appliances	7	23245	Suji appliances
	7	Jeans	Women dress	4	64728	Pappu Readymades
	8	Watch	Accessories	10	75296	Watch Mechanics
	9	Laptop table	Home appliances	9	862263	Suji appliances
	10	Extension box	Home appliances	7	975826	Suji appliances
*	NULL	NULL	NULL	NULL	NULL	NULL

1.5 Display products belonging to the Electronics category.

```
select * from stock where category="Electronics";
```

	product_id	product_name	category	quantity	price	supplier
▶	1	Laptop	Electronics	1	2402900	ABC Electronics
	2	Mobile phone	Electronics	2	23920	Oppo
*	NULL	NULL	NULL	NULL	NULL	NULL

2. Order By:

2.1 Display products in ascending order of price.

```
select * from stock order by price asc;
```

	product_id	product_name	category	quantity	price	supplier
▶	3	Portable Mixer	Home appliances	4	2239	Suji appliances
	4	Short kurti	Women's dress	10	9829	Pappu Readymades
	6	Fan	Home appliances	7	23245	Suji appliances
	2	Mobile phone	Electronics	2	23920	Oppo
	7	Jeans	Women dress	4	64728	Pappu Readymades
	8	Watch	Accessories	10	75296	Watch Mechanics
	5	Chair	Home appliances	8	82749	Suji appliances
	9	Laptop table	Home appliances	9	862263	Suji appliances
	10	Extension box	Home appliances	7	975826	Suji appliances
	1	Laptop	Electronics	1	2402900	ABC Electronics
*	NULL	NULL	NULL	NULL	NULL	NULL

2.2 Display products in descending order of price

```
select * from stock order by price desc;
```

	product_id	product_name	category	quantity	price	supplier
▶	1	Laptop	Electronics	1	2402900	ABC Electronics
	10	Extension box	Home appliances	7	975826	Suji appliances
	9	Laptop table	Home appliances	9	862263	Suji appliances
	5	Chair	Home appliances	8	82749	Suji appliances
	8	Watch	Accessories	10	75296	Watch Mechanics
	7	Jeans	Women dress	4	64728	Pappu Readymades
	2	Mobile phone	Electronics	2	23920	Oppo
	6	Fan	Home appliances	7	23245	Suji appliances
	4	Short kurti	Women's dress	10	9829	Pappu Readymades
	3	Portable Mixer	Home appliances	4	2239	Suji appliances
*	NULL	NULL	NULL	NULL	NULL	NULL

2.3 Display products in ascending order of quantity.

```
select * from stock order by quantity asc;
```

	product_id	product_name	category	quantity	price	supplier
▶	1	Laptop	Electronics	1	2402900	ABC Electronics
	2	Mobile phone	Electronics	2	23920	Oppo
	3	Portable Mixer	Home appliances	4	2239	Suji appliances
	7	Jeans	Women dress	4	64728	Pappu Readymades
	6	Fan	Home appliances	7	23245	Suji appliances
	10	Extension box	Home appliances	7	975826	Suji appliances
	5	Chair	Home appliances	8	82749	Suji appliances
	9	Laptop table	Home appliances	9	862263	Suji appliances
	4	Short kurti	Women's dress	10	9829	Pappu Readymades
	8	Watch	Accessories	10	75296	Watch Mechanics
*	NULL	NULL	NULL	NULL	NULL	NULL

2.4 Display products by category in ascending order and price in descending order.

```
select * from stock order by category asc,price desc;
```

	product_id	product_name	category	quantity	price	supplier
	8	Watch	Accessories	10	75296	Watch Mechanics
	1	Laptop	Electronics	1	2402900	ABC Electronics
	2	Mobile phone	Electronics	2	23920	Oppo
	10	Extension box	Home appliances	7	975826	Suji appliances
	9	Laptop table	Home appliances	9	862263	Suji appliances
	5	Chair	Home appliances	8	82749	Suji appliances
	6	Fan	Home appliances	7	23245	Suji appliances
	3	Portable Mixer	Home appliances	4	2239	Suji appliances
	7	Jeans	Women dress	4	64728	Pappu Readymades
	4	Short kurti	Women's dress	10	9829	Pappu Readymades
	NULL	NULL	NULL	NULL	NULL	NULL

2.5 Display the 5 most expensive products.

```
select * from stock order by price desc limit 5;
```

	product_id	product_name	category	quantity	price	supplier
▶	1	Laptop	Electronics	1	2402900	ABC Electronics
	10	Extension box	Home appliances	7	975826	Suji appliances
	9	Laptop table	Home appliances	9	862263	Suji appliances
	5	Chair	Home appliances	8	82749	Suji appliances
	8	Watch	Accessories	10	75296	Watch Mechanics
*	NULL	NULL	NULL	NULL	NULL	NULL

3. Aggregate Functions:

3.1 Find the total number of products.

```
select count(*) as total_product from stock;
```

	total_product
▶	10

3.2 Find the total available quantity.

```
select count(*) as total_quantity from stock where quantity>0;
```

	total_quantity
▶	10

3.3 Find the average product price.

```
select avg(price) as avg_price from stock;
```

	avg_price
▶	452299.5000

3.4 Find the highest product price.

```
select max(price) as highest_product_price from stock;
```

	highest_product_price
▶	2402900

3.5 Find the lowest product price.

```
select min(price) as lowest_product_price from stock;
```

	lowest_product_price
▶	2239

4. Group By:

4.1 Find the number of products in each category.

```
select category ,count(*) as total_quantity from stock group by category;
```

	category	total_quantity
▶	Electronics	2
	Home appliances	5
	Women's dress	1
	Women dress	1
	Accessories	1

4.2 Find the total quantity available in each category.

```
select category, sum(quantity) as total_quantity from stock group by category;
```

	category	total_quantity
▶	Electronics	3
	Home appliances	35
	Women's dress	10
	Women dress	4
	Accessories	10

4.3 Find the average price of products in each category.

```
select category, avg(price) as avg_price from stock group by category;
```

	category	avg_price
▶	Electronics	1213410.0000
	Home appliances	389264.4000
	Women's dress	9829.0000
	Women dress	64728.0000
	Accessories	75296.0000

4.4 Find the highest-priced product in each category.

```
select category, max(price) as maximum_price from stock group by category;
```

	category	maximum_price
▶	Electronics	2402900
	Home appliances	975826
	Women's dress	9829
	Women dress	64728
	Accessories	75296

4.5 Find the total number of products supplied by each supplier.

```
select supplier, sum(quantity) as total_products from stock group by supplier;
```

	supplier	total_products
▶	ABC Electronics	1
	Oppo	2
	Suji appliances	35
	Pappu Readymades	14
	Watch Mechanics	10

5. Having:

5.1 Display categories having more than 5 products.

```
select category , count(*) as total_quantity from stock group by category having count(*)>=5;
```

	category	total_quantity
▶	Home appliances	5

5.2 Display categories where total quantity is greater than 100.

```
select category, sum(quantity) as total_quantity from stock group by category having sum(quantity)>100;
```

	category	total_quantity
--	----------	----------------

5.3 Display categories where average price is greater than 1000.

```
select category, avg(price) as avg_price from stock group by category having avg(price)>1000;
```

	category	avg_price
▶	Electronics	55185.5000
	Home appliances	428202.0000
	Women's dress	10823.0000
	Women dress	71212.0000
	Accessories	82837.0000

5.4 Display suppliers supplying more than 3 products.

```
select supplier, count(*) as total_quantity from stock group by supplier having count(*)>3;
```

	supplier	total_quantity
▶	Suji appliances	5

5.5 Display categories where the highest product price is greater than 5000.

```
select category, max(price) as max_price from stock group by category having max(price)>5000;
```

	category	max_price
▶	Electronics	84048
	Home appliances	1073420
	Women's dress	10823
	Women dress	71212
	Accessories	82837

6. Update:

6.1 Update the price of a particular product.

```
update stock set price=76397 where product_id=1;
```

```
select * from stock;
```

	product_id	product_name	category	quantity	price	supplier
▶	1	Laptop	Electronics	1	76397	ABC Electronics
	2	Mobile phone	Electronics	2	23920	Oppo
	3	Portable Mixer	Home appliances	4	2239	Suji appliances
	4	Short kurti	Women's dress	10	9829	Pappu Readymades
	5	Chair	Home appliances	8	82749	Suji appliances
	6	Fan	Home appliances	7	23245	Suji appliances
	7	Jeans	Women dress	4	64728	Pappu Readymades
	8	Watch	Accessories	10	75296	Watch Mechanics
	9	Laptop table	Home appliances	9	862263	Suji appliances
	10	Extension box	Home appliances	7	975826	Suji appliances
*	NULL	NULL	NULL	NULL	NULL	NULL

6.2 Increase the price of all products by 10%.

```
update stock set price=price*1.1 where product_id>0;
```

```
select * from stock;
```

	product_id	product_name	category	quantity	price	supplier
▶	1	Laptop	Electronics	1	84048	ABC Electronics
	2	Mobile phone	Electronics	2	26323	Oppo
	3	Portable Mixer	Home appliances	4	2474	Suji appliances
	4	Short kurti	Women's dress	10	10823	Pappu Readymades
	5	Chair	Home appliances	8	91035	Suji appliances
	6	Fan	Home appliances	7	25581	Suji appliances
	7	Jeans	Women dress	4	71212	Pappu Readymades
	8	Watch	Accessories	10	82837	Watch Mechanics
	9	Laptop table	Home appliances	9	948500	Suji appliances
	10	Extension box	Home appliances	7	1073420	Suji appliances
*	NULL	NULL	NULL	NULL	NULL	NULL

6.3 Increase the quantity of a particular product after receiving new stock.

```
update stock set quantity=5 where product_id=7;
select * from stock;
```

	product_id	product_name	category	quantity	price	supplier
▶	1	Laptop	Electronics	1	84048	ABC Electronics
	2	Mobile phone	Electronics	2	26323	Oppo
	3	Portable Mixer	Home appliances	4	2474	Suji appliances
	4	Short kurti	Women's dress	10	10823	Pappu Readymades
	5	Chair	Home appliances	8	91035	Suji appliances
	6	Fan	Home appliances	7	25581	Suji appliances
	7	Jeans	Women dress	5	71212	Pappu Readymades
	8	Watch	Accessories	10	82837	Watch Mechanics
	9	Laptop table	Home appliances	9	948500	Suji appliances
	10	Extension box	Home appliances	7	1073420	Suji appliances
*	NULL	NULL	NULL	NULL	NULL	NULL

6.4 Change the supplier of a particular product.

```
update stock set supplier="Arjun appliances" where product_id=5;
select * from stock;
```

	product_id	product_name	category	quantity	price	supplier
▶	1	Laptop	Electronics	1	84048	ABC Electronics
	2	Mobile phone	Electronics	2	26323	Oppo
	3	Portable Mixer	Home appliances	4	2474	Suji appliances
	4	Short kurti	Women's dress	10	10823	Pappu Readymades
	5	Chair	Home appliances	8	91035	Arjun appliances
	6	Fan	Home appliances	7	25581	Suji appliances
	7	Jeans	Women dress	5	71212	Pappu Readymades
	8	Watch	Accessories	10	82837	Watch Mechanics
	9	Laptop table	Home appliances	9	948500	Suji appliances
	10	Extension box	Home appliances	7	1073420	Suji appliances
*	NULL	NULL	NULL	NULL	NULL	NULL

7. Delete:

7.1 Delete a product using its product_id.

```
delete from stock where product_id=10;  
select * from stock;
```

	product_id	product_name	category	quantity	price	supplier
▶	1	Laptop	Electronics	1	84048	ABC Electronics
	2	Mobile phone	Electronics	2	26323	Oppo
	3	Portable Mixer	Home appliances	4	2474	Suji appliances
	4	Short kurti	Women's dress	10	10823	Pappu Readymades
	5	Chair	Home appliances	8	91035	Arjun appliances
	6	Fan	Home appliances	7	25581	Suji appliances
	7	Jeans	Women dress	5	71212	Pappu Readymades
	8	Watch	Accessories	10	82837	Watch Mechanics
	9	Laptop table	Home appliances	9	948500	Suji appliances
*	NULL	NULL	NULL	NULL	NULL	NULL

7.2 Delete products whose quantity is 0.

```
delete from stock where quantity=0;  
SET SQL_SAFE_UPDATES = 0;  
select * from stock;
```

	product_id	product_name	category	quantity	price	supplier
▶	1	Laptop	Electronics	1	84048	ABC Electronics
	2	Mobile phone	Electronics	2	26323	Oppo
	3	Portable Mixer	Home appliances	4	2474	Suji appliances
	4	Short kurti	Women's dress	10	10823	Pappu Readymades
	5	Chair	Home appliances	8	91035	Arjun appliances
	6	Fan	Home appliances	7	25581	Suji appliances
	7	Jeans	Women dress	5	71212	Pappu Readymades
	8	Watch	Accessories	10	82837	Watch Mechanics
	9	Laptop table	Home appliances	9	948500	Suji appliances
*	NULL	NULL	NULL	NULL	NULL	NULL

7.3 Delete all products from a particular category.

```
53 • delete from stock where category="Accessories";
54 • select * from stock;
```

Result Grid						
		Filter Rows:	Edit:		Export/Import:	
	product_id	product_name	category	quantity	price	supplier
▶	1	Laptop	Electronics	1	84048	ABC Electronics
	2	Mobile phone	Electronics	2	26323	Oppo
	3	Portable Mixer	Home appliances	4	2474	Suji appliances
	4	Short kurti	Women's dress	10	10823	Pappu Readymades
	5	Chair	Home appliances	8	91035	Arjun appliances
	6	Fan	Home appliances	7	25581	Suji appliances
	7	Jeans	Women dress	5	71212	Pappu Readymades
	9	Laptop table	Home appliances	9	948500	Suji appliances
*	NULL	NULL	NULL	NULL	NULL	NULL

7.4 Delete products supplied by a particular supplier.

```
55 • delete from stock where supplier="Oppo";
56 • select* from stock;
```

Result Grid						
		Filter Rows:	Edit:		Export/Import:	
	product_id	product_name	category	quantity	price	supplier
▶	1	Laptop	Electronics	1	84048	ABC Electronics
	3	Portable Mixer	Home appliances	4	2474	Suji appliances
	4	Short kurti	Women's dress	10	10823	Pappu Readymades
	5	Chair	Home appliances	8	91035	Arjun appliances
	6	Fan	Home appliances	7	25581	Suji appliances
	7	Jeans	Women dress	5	71212	Pappu Readymades
	9	Laptop table	Home appliances	9	948500	Suji appliances
*	NULL	NULL	NULL	NULL	NULL	NULL

7.5 Delete products whose price is less than 100.

```
delete from stock where price<100;
select * from stock;
```

	product_id	product_name	category	quantity	price	supplier
▶	1	Laptop	Electronics	1	84048	ABC Electronics
	3	Portable Mixer	Home appliances	4	2474	Suji appliances
	4	Short kurti	Women's dress	10	10823	Pappu Readymades
	5	Chair	Home appliances	8	91035	Arjun appliances
	6	Fan	Home appliances	7	25581	Suji appliances
	7	Jeans	Women dress	5	71212	Pappu Readymades
	9	Laptop table	Home appliances	9	948500	Suji appliances
*	NULL	NULL	NULL	NULL	NULL	NULL

8. Stock Condition Queries:

8.4 Display products whose quantity is greater than 100.

```
59 • select product_name, quantity from stock where quantity>100;
```

Result Grid			Filter Rows:		Export:		Wrap Cell Content:	
product_name	quantity							

8.5 Display products whose quantity is between 10 and 50.

```
60 • select product_name, quantity from stock where quantity >10 and quantity<50;
```

Result Grid			Filter Rows:		Export:		Wrap Cell Content:	
product_name	quantity							

9. Calculated Columns:

9.1 Display the product name and total stock value

```
61 • select product_id,product_name, quantity*price as stock_value from stock;
```

Result Grid			Filter Rows:		Export:		Wrap Cell Content:	
product_id	product_name	stock_value						
1	Laptop	84048						
3	Portable Mixer	9896						
4	Short kurti	108230						
5	Chair	728280						
6	Fan	179067						
7	Jeans	356060						
9	Laptop table	8536500						

9.2 Display products with a 10% increased price.

```
62 • select product_id, product_name , price, price*1.1 as percentage_increase from stock;
```

Result Grid			Filter Rows:		Export:		Wrap Cell Content:	
product_id	product_name	price	percentage_increase					
1	Laptop	84048	92452.8					
3	Portable Mixer	2474	2721.4					
4	Short kurti	10823	11905.3					
5	Chair	91035	100138.5					
6	Fan	25581	28139.1					
7	Jeans	71212	78333.2					
9	Laptop table	948500	1043350.0					

9.3 Display the total value of all available stock.

```
63 • select product_name,sum(quantity*price) as available_stock from stock group by product_name;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	product_name	available_stock		
▶	Laptop	84048		
	Portable Mixer	9896		
	Short kurti	108230		
	Chair	728280		
	Fan	179067		
	Jeans	356060		
	Laptop table	8536500		

9.4 Display product name, quantity, price, and stock value.

```
64 • select product_name, quantity, price, quantity*price as stock_value from stock;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	product_name	quantity	price	stock_value
▶	Laptop	1	84048	84048
	Portable Mixer	4	2474	9896
	Short kurti	10	10823	108230
	Chair	8	91035	728280
	Fan	7	25581	179067
	Jeans	5	71212	356060
	Laptop table	9	948500	8536500

9.5 Display products whose total stock value is greater than 50,000.

```
65 • select product_id,product_name, quantity*price as stock_value from stock where quantity*price>50000;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	product_id	product_name	stock_value	
▶	1	Laptop	84048	
	4	Short kurti	108230	
	5	Chair	728280	
	6	Fan	179067	
	7	Jeans	356060	
	9	Laptop table	8536500	

